Detailed Project Report

RRBO

ULO Re- Refining by distillation & hydro treatment plant

Installed Capacity of 65 TPD

A Venture of SANTOSH PETROCHEMICAL INNOVATIONS PVT LTD which is part of Group Business of SANTOSH ASSOCIATES

PROFILE

Santosh Petrochemical Innovations PVT LTD is a group entity of Santosh Associates. Santosh Associates appointed as Servo Stockiest Industrial (SSI) under Indian Oil's Business Process Restructuring in the year 2003 & is to sell SERVO lubricants for six districts of Western U.P. Industrial belt, viz Ghaziabad, Meerut, Muzaffarnagar, Baghpat, Shamli, Saharanpur.

Santosh Associates-LPG Section launched in December 2018 for LPG Bottling at Amroha-UP as a private bottlers on behalf of Hindustan Petroleum Corporation Limited. We are bottling 30TMT per annum in cylinders for domestic and commercial purposes and distributing these with the accordance of HPCL for their dealers and distributors for Amroha region.

Recently, Santosh Associates have been awarded LOA for Operations & Maintenance of HPCL Sitarganj LPG Plant for 5 years under third party operations scheme. Total Quantity of LPG Bottling shall be of 340TMT in entire contract tenure.

PROPOSED PROJECT - ULO Re- Refining by distillation & hydro treatment plant

Santosh Petrochemical Innovations PVT LTD is planning to establish a Plant for ULO Re- Refining by distillation & hydro treatment plant for treatment of LOBS.

This Plant shall be established with a installed capacity of 200 TPD Feed Capacity USED OIL RE-REFINERY Facility for recovering LOBS(Lube Oil Base Stock) from used Lubricating Oils and to Hydro-Treat it to produce API-Gr-II+ Oils under flagship- "Santosh Petrochemical Innovations PVT LTD" with the technological solution and guidance of REVA PROCESS TECHNOLOGIES P. LTD, PUNE & European country.

Such a plant will be situated in the western part of Uttar Pradesh. Basic technical Specification with Financial Projection is detailed in this Report

This re-refined base oil (RRBO) can then be used again to produce new lubricants, helping to reduce waste and emissions associated with traditional motor oil production process. And, as per guide lines of Government of India, under EPR (Extended Producer Responsibility) policy, the lubricant manufactures to reduce waste, use less packaging, and develop more sustainable products. It can also contribute to a circular economy by encouraging reuse, buy back, and recycling programs. This year government fixed the target to use 5% RRBO for lubricant manufactures, then after, to add 5% on yearly basis up to sealing 50% of RRBO.

PROMOTERS

a. Lalit Bindal
b. Robin Kumar
c. Pooja Bindal
d. Abhinav Bindal

Name	Lalit Bindal
Father Name	Guru Charan Bindal
DOB	01st Oct 1966
Residential Address	R-14/84, Raj Nagar, Ghaziabad-201002
PAN No.	AATPL9346F
Aadhar No.	7574 4586 1090
Experience	-Proprietor of "SANTOSH ASSOCIATES" -Santosh Associates is a SSI-Indian Oil Corporation Limited that deals in trading and marketing of Servo Brand lubricants in 6 districts of western UP. Having private bottling assistance to HPCL at Amroha Having O&M services of LPG Bottling Plant of HPCL at Starganj. Having huge experience of 35 years put altogether.

Name	Robin Kumar
Father Name	Harveer Singh
DOB	16th July 1984
Residential Address	D-113, Sector-61, Noida, Gautam Budh Nagar, UP-201301
PAN No.	AVAPK8696D
Aadhar No.	3051 1606 6710
Experience	

Name	Abhinav Bindal
Father Name	Lalit Bindal
DOB	22nd February 1995
Residential Address	R-14/84, Raj Nagar, Ghaziabad-201002
PAN No.	CGBPB7157E
Aadhar No.	9464 7019 4530
Experience	-Working with Nabih Youseef Associates as Structural Engineer & Analyst of earth quake, Los Angles United State

Name	Pooja Bindal
Husband Name	Lalit Bindal
DOB	19th December 1967
Residential Address	R-14/84, Raj Nagar, Ghaziabad-201002
PAN No.	ABVPB4203A
Aadhar No.	9807 6149 7618
Experience	-Proprietor of "BINDAL CREATION" - BINDAL CREATION that deals in trading of traditional and western garments outfit at Ghaziabad. -She has experience of 8 years in this industry.

TEV REPORT

This Techno-Economic Viability (TEV) report assesses the feasibility of establishing a used oil re-refining project in India, focusing on converting waste lubricating oils into high-quality base oils. The project aligns with India's push towards a circular economy, reducing crude oil imports, and complying with environmental regulations.

The purpose of this TEV is to evaluate technical and economic factors for risk mitigation, informed decision-making, and overall feasibility assessment. Lenders and investors can use it to gauge potential returns while addressing sustainability goals. Key benefits include conserving resources, lowering emissions (re-refining uses 66% less energy than virgin oil production), and supporting Extended Producer Responsibility (EPR) mandates under the Hazardous and Other Wastes (Management and Transboundary Movement) Second Amendment Rules, 2023.

1. MARKET ANALYSIS

The used oil recycling market in India primarily refers to the management, collection, and processing (including re-refining) of used lubricating oil (also called waste engine oil, spent oil, or used motor oil) from automotive, industrial, and other sources. This is distinct from used cooking oil (UCO), which is often collected for biodiesel production and has its own separate market dynamics.

- **Current Market Overview**

India generates a substantial volume of used lubricating oil due to rapid industrialization, growing vehicle ownership, and expanding manufacturing sectors. Estimates indicate over 1.3 million metric tons (MMT) of used lubricating oil generated annually as of recent years (around 2024–2025), with projections for steady increases tied to lubricant consumption growth.

However, formal recycling remains limited:

- Less than 15% is formally recycled.
- Much of the rest enters informal channels, often burned as low-grade fuel, dumped, or mismanaged, posing environmental and health risks (e.g., soil/water contamination and emissions).

The lubricant market (which drives used oil generation) stood at around 4.2 MMT in FY22 and is projected to grow to 4.8–5.6 MMT by FY29 or 5.0–6.0 MMT by FY31. This implies rising used oil volumes, creating feedstock potential for recycling.

Re-refining used oil into high-quality base oils or lubricants offers benefits like reduced crude oil imports, lower energy use, and GHG emissions savings, supporting India's circular economy and self-reliance goals.

- **Regulatory Framework**

The sector is governed by the Hazardous and Other Wastes (Management and Transboundary Movement) Rules, 2016 (amended over time), classifying used oil as hazardous waste. Key developments include the Extended Producer Responsibility (EPR) framework for used oil, effective from April 1, 2024 (notified under these rules).

- Producers/importers of lubricants must meet recycling targets via EPR certificates from registered recyclers.
- Targets start at 5% of oil sold in FY2023 for FY2025, scaling to 50% by FY2031.
- This is managed digitally via the Central Pollution Control Board (CPCB) portal.
- The shift aims to formalize collection, reduce informal diversion, and boost registered re-refining.

Reports (e.g., FICCI-RECEIC-Deloitte 2025) highlight the need for phased reforms, technology upgrades, track-and-trace systems, and integration of informal sectors to meet these targets effectively.

- **Market Size and Growth**

Specific India-only figures for used lubricating oil recycling are emerging but fragmented, with the sector transitioning from informal to organized.

- Broader waste oil recycling (including used lubricating oil) in India is estimated at around USD 3.38 billion in 2025 (from broader Asia-Pacific breakdowns), with strong growth.
- One analysis projects India's waste oil recycling market revenue growing from ~USD 2.5 billion (earlier base) to USD 6.22 billion by 2033, at a CAGR of ~7.9%.
- Another view pegs the waste oil recycling industry (with engine oil focus) at USD 4.05 billion in 2025, rising to USD 6.37 billion by 2033.
- Globally, recycled/re-refined base oil markets show Asia-Pacific (including India) leading growth, with India's contribution rising due to policy push.

The re-refined base oil segment benefits from EPR mandates and investments in advanced technologies (e.g., hydroprocessing for higher-quality outputs).

Key Drivers and Opportunities

- Rising lubricant/vehicle demand — More used oil feedstock.
- EPR compliance — Forces organized collection and boosts formal recycling volumes.
- Environmental and economic benefits — Re-refining saves resources; potential for base oils, fuels, or other uses.
- Government initiatives — Circular economy focus, import reduction, and incentives for re-refiners.
- Investments — Upgrades in re-refining plants and collection networks.

Overall, India's used oil recycling market is at a transformative stage in 2026, driven by EPR regulations and sustainability goals. It holds strong growth potential (likely 6–8%+ CAGR in coming years), shifting toward formal, circular models that could recycle 0.85–1.00 MMT annually by FY2031 under targets. This positions the sector as key to reducing environmental impact while creating economic value.

2. TECHNICAL FEASIBILITY

Used oil re-refining in India is technically viable with mature technologies adapted to local feedstock variability. Processes yield 70-75% base oils, compliant with BIS standards.

➢ **Key Technologies**

- **Vacuum Distillation with Hydrotreating**: Removes impurities under vacuum; yields Group II/III oils. Widely used for high recovery (70-75%).
- **Thin-Film Evaporation/Short Path Distillation**: Efficient for asphaltene removal; minimizes thermal degradation.
- **Solvent Extraction/Acid-Clay Treatment**: Removes contaminants; suitable for small-scale operations.

➢ **Infrastructure and Operations**

- Feedstock: Used motor, industrial oils; collection via EPR networks.
- Process Steps: Pre-treatment (filtration, dehydration), distillation, contaminant removal, blending.
- Outputs: Base oils (Group I/II), diesel, asphalt extender.
- Scalability: Viable for 200 TPD; integrates with existing facilities.
- Challenges: like feedstock quality mitigated by testing and EPR traceability.

Technical feasibility is high, supported by 257 facilities, but requires skilled labor and CPCB compliance.

- **Process Description**

The important step in re-refining of used oil is selection of feed stock, local rules & regulation may not allow processing used oil with high total halogens (high PCB's content). Next important step in process is sedimentation & filtration of feed stock. Correct filtration process will ensure smooth running of plant without fouling & coking. Pre-screened feed shall be filtered down to 150 - 250-micron level by basket filters. These filters shall be installed in tank farm.

The heater installed will heat feed material to 120 Deg. C where almost all water will come out of used oil. A Heat Recuperator installed in between will recover heat from the process. This dry oil is ready for next step of purification. We recommend chemical injection in process to avoid fouling due to additives present in used oil. This filtered & dried used oil will be sent to next step where almost all glycol, solvent and light ends present in used oil will be separated. This process occurs under vacuum. Solvents & light fuel ends condensed at overhead condenser will be separated in specially designed Oil - Water Separator (OWS) & will be sent to storage separately. This recovered light fuel in some cases can be used as plant fuel or it can be sold as fuel in market.

Thus dry / dehydrated oil will be sent for next step i.e. distillation to separate lube oil from additives & high boiling hydrocarbons. Dry used oil is heated further to 280 Deg. C. – 300 Deg C. The distillation of the oil will be done in 2 steps. First medium density oil is received via a distillation column and then bottom of distillation column will be passed through an evaporator system to recover heavy distillate oil.

A Wiped Film Evaporator will be used at the end of process to recover ultra heavy material which operates at higher vacuum & higher temperatures 300 Deg. C. – 310 Deg. C. This will ensure the reliability of process. This step will recover almost all remaining LOBS & spent additives are removed from bottom of WFE. This bottom by-product is Asphalt Residue & is a saleable product as asphalt extender or in tar applications. Depending on feed stock characteristics a light distillate cut can also be recovered during distillation process.

The waste water generated at dehydration (OWS system) step will be sent to effluent treatment facility for disposal.

Distillate recovered in evaporation section will be sent to hydrotreatment process, hydro treating process involves reaction of hydrogen-rich gas with oil, in this case distillate oil from used lubricating oil (ULO - LOBS), under calculated conditions of pressure and temperature in presence of proprietary metallic catalysts.

The reactions include;

a) Reactions with chlorine, sulfur, nitrogen that produce inorganic products such as HCl, H2S and NH3.

b) Saturation of olefins that produce a stable, oxidation resistant product

c) Aromatics saturation.

After the reactions are carried out in specially designed reactors, the hydrogen-rich gas and oil are separated in a flash drum. The high-pressure gas is neutralized, cooled and recycled back to the reactors. Make-up hydrogen is added to compensate for reaction losses. The oil under high pressure is depressurized and sent to the stripping section. When oil is depressurized, a vapor stream is created and purged for disposal. The purged vapor contains H2S and light hydrocarbons. The purged stream can be directly burned in thermal oil heater for heat recovery or sent to a thermal oxidizer for disposal.

Wastewater from the process is contaminated with free-floating

oil. The water is sour because of dissolved hydrogen sulfide, ammonia and phenol. Free-floating oil is separated from wastewater in a specially designed coalescing separator. The oil-free sour-water is heated and stripped with low-pressure steam. As a result, the concentration of H2S, ammonia and phenols in sour-water is substantially reduced. The wastewater is then sent for further treatment in the wastewater treatment plant.

Thermal oil system provides heat to the Process using hot thermal oil. The thermal oil after transferring heat to the Process Plant is looped back to the Thermal Oil System for reheating. The system consists of the furnace (or heater), the thermal oil circulation pumps, burners, air blowers, an expansion tank to accommodate thermal expansion of thermal fluid and the interconnecting piping. A control system detects the flame, manipulates flow of air to the furnace and controls oxygen content in the flue gas to ensure complete combustion. The burners are capable of both liquid and gaseous fuels as well as process off-gas.

Steam is used in the Plant for stripping, heating and tank-heating. Steam is also needed at utility stations for cleaning and flushing of pipelines and equipment during maintenance as well as operations. The steam generation system consists of the steam-generating heat exchanger, a deaerator / steam drum, feed pumps. Steam is generated by heating softened water in a heat exchanger. The heat necessary for generation of steam is produced by thermal oil (heat transfer fluid). Only a part of the steam condensate is returned to the generator. A fraction of water in the steam-drum is drained to control conductivity of water. The steam lost in the process and the drain is made up with supply of soft water to the steam generator. The generator and steam piping require chemical treatment to prevent corrosion, and scaling. The chemicals and the dosing pumps can be supplied by an industrial chemical supplier on a contract.

The cooling water circulation system removes heat from the Process Plant using water. The “cooling water” in turn gets heated in the Process Plant and looped back to the Cooling System. The Cooling System will cool the water by evaporating a small part of the cooling water. An additional small part is drained for maintaining dissolved solids content of cooling water. The evaporation and drain loss is made up with fresh water supply continuously. The system consists of the cooling tower, two circulation pumps and the interconnecting piping. The cooling water loop requires chemical treatment to prevent corrosion, scaling and prevent bio-fouling of the piping, heat exchangers in the Process Plant. The chemicals and the dosing pumps are supplied by an industrial chemical supplier on a contract.

The air compressor system provides dried filtered air for operation of control valves and other pneumatic equipment in the Process Plant. The system consists of an air-compressor, set of filters and driers and a hold-up tank with interconnecting piping and control panel.

The seal flushing system provides oil-flow for flushing of bearing and mechanical seals on rotating equipment in the Process Plant. The oil is circulated from the system to the Process Plant and back to the seal oil tank in a loop. The system consists of the tank, the circulation pumps, filter and a cooler to maintain the quality and temperature of seal oil supply to the Process Plant.

A tank fitted with agitators; heating coils is used to store caustic soda (NaOH) usually stored as 42 to 48 % wt. aqueous solution. Caustic soda pumps are used to pressurize a header and supply system. Caustic solution is used in the process plant for acid neutralization.

A. PROCESS PLANT SECTION SUPPLY:

A.1. DISTILLATION SECTION:

Process Plant Area			
Sr. No.	**Section**	**Purpose**	**Main Components**
1.	Dehydration	To remove water, glycol, light ends / fuel from wet used motor oil.	Dehydration Column, Pumps (Operational + Spare), Condensers.
2.	Vacuum Distillation	To Separate Lube Oil Base Stock from Dry Used oil. WFE will be used.	Evaporators, Column.
			Condensers.
			Pumps with Double Mechanical Seals (Operating + Spare).
			Asphalt Residue Collector.
			Traps for entrainment removal.
			Coolers required for vacuum System.
			Sub-coolers.
			Piping.
3.	Vacuum system, Sub-cooler.	To maintain Vacuum in the system	Vacuum Pumps Package.
4.	WFE	WFE with Motor, Internal Condenser and rotor dynamically balanced.	BY REVA.
5.	Motor for Pumps	Motor for pumps are considered considering that the pumps are with mechanical seal.	BY REVA.

A.2. HYDROTREATMENT SECTION

B. Process Plant Area			
Sr. No	**Section**	**Purpose**	**Main Components**
1.	Hydro treatment	To remove Sulfur, Metals and aromatic compound from LOBS lube oil recovered from Used Lubricating Oil.	All Equipment's required for Plants, Reactors – 3 Nos. with Catalyst Required for process (First Fill). Heaters, Heat Exchangers, Heat Recuperators, Condensers, Coolers, Pressure Vessels, Hold-Up Vessels, Pumps, Recycle Hydrogen Compressor, Stripping System, Oil-Water Separator, Waste water stripper.
2.	Vacuum system, Sub-cooler.	To maintain Vacuum in the system.	BY REVA
3.	Motor for Pumps	Motor for pumps are considered considering that the pumps are with mechanical seal. REVA TO modify the pumps to suit the process requirement.	BY REVA

B.1. FRACTIONATION SECTION:

C. Process Plant Area			
Sr. No	**Section**	**Purpose**	**Main Components**
1.	Fractionation	To separate different distillate cuts in-order to recover market ready cuts from LOBS.	All Equipment's required for Plants, Fractionation Column Pre- Heaters, Heat Exchangers, Heat Recuperators, Condensers, Column, Column Internal, Coolers, Pressure Vessels, Hold-Up Vessels, Pumps.
2.	Vacuum system, Sub-cooler system	To maintain Vacuum in the system.	BY REVA
3.	Motor for Pumps	Motor for pumps are considered considering that the pumps are with mechanical seal. REVA TO modify the pumps to suit the process requirement.	BY REVA

C.1. TANKFARM & EFFLUENT SYSTEM SECTION:

	Tank Farm		
1	Tank - 1000 KL X 2 Nos	CS	2
2	Tank - 500 KL X 2 Nos	CS	2
3	Tank - 200 KL X 2 Nos	CS	2
4	Tank - 100 KL X 2 Nos	CS	2
5	Tank Farm Pumps	CS	8
6	Filter System For Unloading	CS	1
7	Vapor Recovery System	CS	1
8	ULO Transfer Pump	CS	2
9	Unloading Filter	CS	2
10	Wastewater Pump	CS	1
	Effluent Abatement System		
1	ETP	CS+FRP	1
2	Scrubber	CS+FRP	1
3	Thermal Oxidizer	CS	1
4	Flare System with KO Pot	CS	1

C.2. UTILITIES SECTION

Utility Area			
Sr. No.	**Section**	**Purpose**	**Main Components**
1	Thermal Oil heater package	To provide heat to process	Fired Heater package.
			Thermal oil circulating Pumps (Operating + Standby).
			Expansion & De-Aeration tank with by pass station
2	Cooling Tower	Evaporative cooling tower, to remove heat from plant	Evaporative cooling Tower.
			Circulating pumps
			FRP / GCS Sump & Strainer
3	Seal Fluid System	To provide lubrication to pump mechanical seals.	Seal fluid tank
			Cooler.
			Circulating pumps.
4	Compressed Air System	To provide air to instruments & Utilities.	Air Compressor
			Dryer for oil & water removal
			Storage tank.
5	Chemical Injection Packages	Chemical Injection Packages for 1. Process 2. Soft Water 3. Cooling Tower 4. Steam Generator Process	Chemical Injection Packages

C.3. STRUCTURAL STEEL:

Structural Steel			
Sr. No.	**Section**	**Purpose**	**Main Components**
1.	Structure	To support all equipment's and pipes in process plant and Utility area. Entire structure will be fabricated as per design to suit local codes and will be shipped in dismantled condition.	Designed & fabricated as per local national code. Walkways & Railings required will be shipped loose to be installed at site. Gratings will be shipped loose. REVA will apply 2 Coats of primer paint of Inorganic Zinc Silicate on structure of 85 DFT. Structure required outside plant area and for pipe-rack is in CLIENT Scope. Total 175 Tons of steel structure assumed for the plant and pipe rack. Any additional material required due to site conditions then it will be charged extra at Rs. 145 / Kg.

C.4. CONTROL SYSTEM:

The entire plant is controlled by SCADA and PLC based system, controlling all the process parameter and operation (sequential start and stop) from the computer screen.

The internationally reputed brand of PLC will be used for accessing all the process parameter whereas SCADA (Supervisor control and data Accusation System) will be the front end of the operator/ controlling system. The process parameter can be changed, monitored and analyzed by the SCADA System. SCADA will have the capacity of generating report (trends, history) for different level of controlling authority. With Ethernet capability one can access the data from remote computer, hence even the plant manager can log on the system form the other place and can control the plant operation. There will be no hot redundancy & ESD. The automation system will reduce the requirement of operator to the level of only one per shift with three helpers which reduces the cost of operation as well as variation in quality due to human interferences.

Control System – Plant wide control System (PCC)			
Sr. No.	**Section**	**Purpose**	**Main Components**
1.	Control System for Process, Utilities	Plant wide control system for process & utilities.	Prewired & tested control panel with branded PLC.
			SCADA.
			Operating & Engineering Station.
			Printer.
			UPS for 30 min. back-up to computers.
			Computer with 2 Nos. LCD display.

Control System – MCC			
Sr. No.	**Section**	**Purpose**	**Main Components**
1.	Electrical MCC for Process & Utilities.	MCC	Prewired & tested control panel.
			VFD's Required for Pumps in process & utilities.
			Relays, Switchgear etc.
			Junction Boxes (2 Nos.)
			Push Buttons
			Indicating Lights

C.5. INSTRUMENTS (REVA SCOPE OF SUPPLY):

Instruments			
Sr No	**Section**	**Purpose**	**Main Components**
1.	Process, Utilities	For Analog & Digital process parameters Readings & recordings.	Flow, Level, Pressure & Temperature & pH Instruments.

C.6. PIPE & PIPE FITTINGS (REVA SCOPE OF SUPPLY):

Piping			
Sr. No.	**Section**	**Purpose**	**Main Components**
1.	Process, Utilities	For all Section as listed 4.2. For process & utilities area all piping with valve is considered till process plant battery limits.	Piping in plant, Flanges, Valves and Pipe fittings. Specialized Piston Valves for thermal fluid & High pressure, temperature.

C.7. LAB INSTRUMENTS (REVA SCOPE OF SUPPLY):

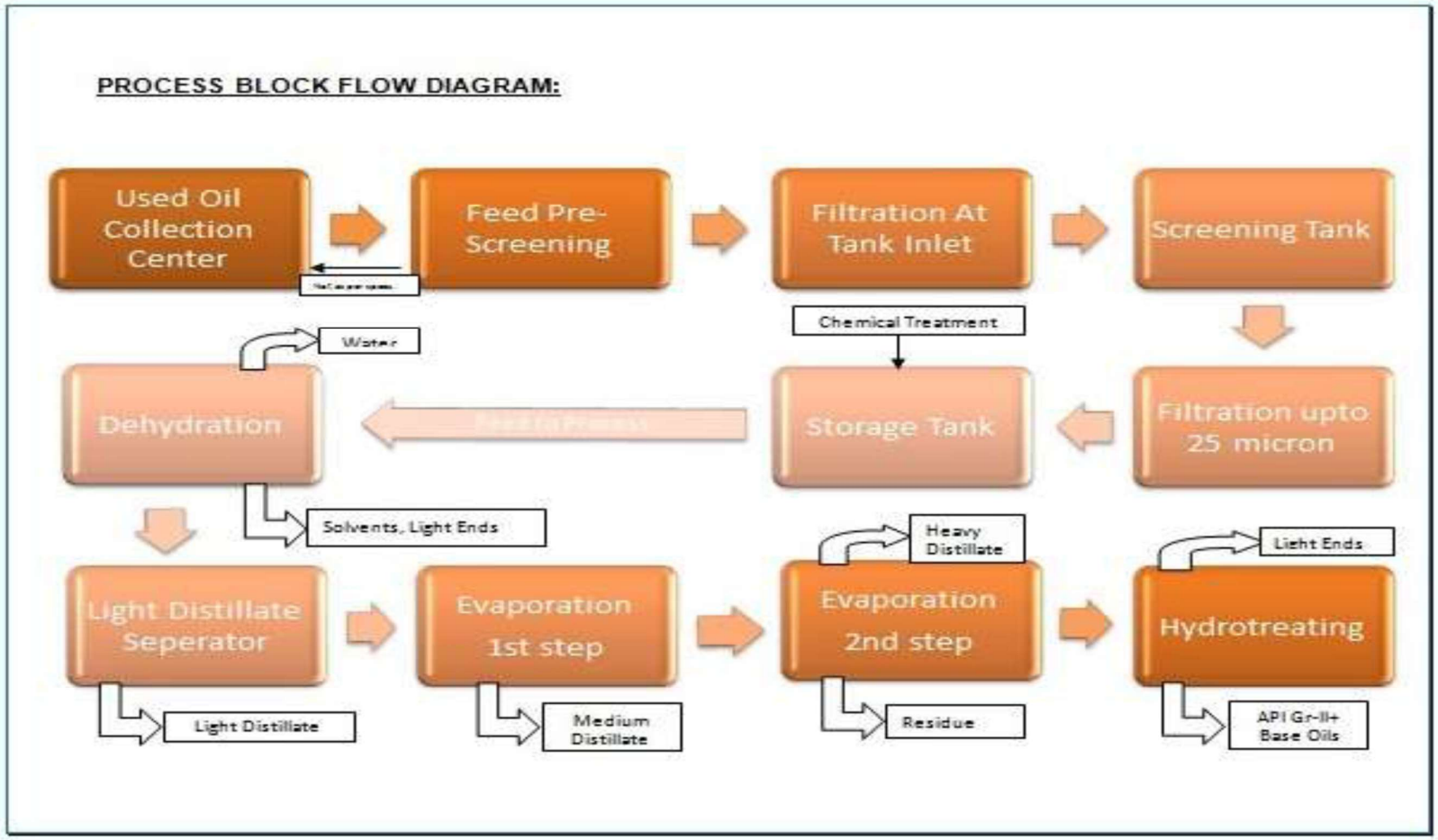
LAB INSTRUMENTS			
Sr. No.	**Type**	**Purpose**	**Qty**
1.	Viscometer	To Measure Viscosity	2
2.	Density	To Measure Density of Oil	2
3.	Karl-Fisher	To Measure Water Content	1
4.	Analytical Balance	For Weight Measurement	1
5.	BOD Incubator	To measure BOD	1
6.	Bomb calorimeter	To Measure Calorific Value	1
7.	Centrifuge	For Centrifugation	1
8.	COD digester	To Measure COD	1
9.	Compressor mini with accessories	Air Compressor	1
10.	Conductivity Meter	To Measure TDS	1
11.	Deep freezer	Freezer	1
12.	DO Meter	To Measure DO	1

13.	Flame Photometer	Photometer	1
14.	Flash point apparatus	To Measure Flash Point	1
15.	Hot Air Oven	Oven	1
16.	Hot Plate	For Heating	1
17.	Jar Test Apparatus	Jar Test	1
18.	Kjeldahl Digestion and Distillation Unit	Distillation	1
19.	Laboratory Stirrer	For Stirring	2
20.	Magnetic Stirrer	Magnetic Stirrer	2
21.	Mechanical Shaker	Shaking	1
22.	Micrometer (digital)	Measurement	1
23.	Filtration unit	For Filtration	2
24.	Muffle furnace	Furnace	1
25.	pH Meter	To Measure pH	1
26.	Refrigerator	-	1
27.	TKN Analyzer	-	1

28.	Turbidity Meter	To Measure Turbidity	1
29.	Water bath	For Heating	1
30.	Fume Hood with scrubber	For Safe Gas Removal	1
31.	Titration Apparatus	-	6
32.	Suction Pump	For Vacuum	1
33.	Glassware	All Glass Ware required	1
34.	Bunsen Burner and Gas with Tripod & Wire mesh	For Heating	2
35.	Petri Dishes	-	12
36.	Distillation Setup	-	1
37.	Blender	-	1
38.	Glass Rods	-	12
39.	Graduated Dropper	-	12
40.	Tongs	-	6
41.	Wash Bottles	-	6

42.	Spatula	-	12
43.	Glass Condensers	-	3
44.	Separator Funnel	-	6
45.	Filtering Flasks	-	6
46.	Mortar and pestle	-	2
47.	Reagent Bottles	-	24
48.	Sample Containers	-	100
49.	Desiccators	-	2

- **Manufacturing Process Flow Chart**



3. RISK ASSESSMENT

Fire and Explosion: Used oil has a flash point, and the process involves heating it to high temperatures (e.g., >200°F or 93°C), creating significant fire risks if not managed.

High-Pressure Hydrogenation/Hydrotreating: The hydrotreating unit uses hydrogen at high pressure, which poses risks of hydrogen embrittlement, leaks, and potential catastrophic failure of reactors if not designed properly.

Chemical Exposure: Used oil contains toxic contaminants, including polycyclic aromatic hydrocarbons (PAHs), lead, cadmium, chromium, and PCBs. Chronic exposure can cause skin irritation, dermatitis, and potential carcinogenicity.

Environmental Pollution: Spills, leakages from storage tanks, pipelines, or during transportation can severely contaminate soil and groundwater. A single gallon of used oil can contaminate 1 million gallons of water.

Hazardous Waste Generation: The process generates acid sludge and spent catalysts (e.g., Mo-Ni/Al), which are highly toxic and require specialized disposal.

Confined Space Hazards: Workers entering tanks or vessels for maintenance are at risk of exposure to volatile organic compounds (VOCs), Hydrogen Sulphide (H2Scap H sub 2 cap S$H2S$), and oxygen-deficient atmospheres.

➢ Risk Mitigation Strategies (Controls)

❖ **Engineering Controls:**

- Implement leak detection systems.
- Design plant with respect to Indian codes and standards.
- Install secondary containment (dyke walls) for all storage tanks.
- Use closed systems for transferring oil to minimize manual handling.
- Ensure proper grounding and bonding of tanks and pipelines to prevent static electricity build-up.
- Equip the plant with proper ventilation, emergency eyewash stations, and safety showers.

❖ **Operational Controls:**

- Strictly follow standard operating procedures (SOPs) and use only licensed transporters.
- Store used oil for a limited time (e.g., minimum 15 days).
- Implement a rigorous, weekly, or regular training program for employees on hazard identification.
- Maintain detailed records of incoming/outgoing oil for traceability.

❖ **Personal Protective Equipment (PPE):**

- Require PPE such as chemical-resistant gloves (nitrile, neoprene), safety goggles, and respiratory protection (especially when dealing with vapours or mists).
- Use heat-resistant gear when handling heated oils.

❖ **Emergency Response:**

- Install fire detection and firefighting systems (foam, dry chemical).
- Have a documented, tested emergency evacuation and disaster management plan.

❖ **Regulatory Compliance**

Used oil is classified as hazardous waste, necessitating compliance with regulations like the Hazardous and Other Wastes (Management and Transboundary Movement) Rules, 2016 (or equivalent local regulations). This includes obtaining proper authorization from environmental agencies, managing spill containment systems, and ensuring proper disposal of generated toxic waste.

4. SWOT Analysis

➢ Strengths (Internal)

- **High-Quality Output:** Hydro treatment produces high-quality API Group II/III base oils (65–80% yield) that can match or exceed the performance of virgin base oils, allowing for high-value sales.
- **Sustainable and Environmental Friendly:** The process converts hazardous waste into valuable resources, reduces dependency on crude oil imports, and significantly lowers CO_2 emissions (up to 50% less energy required compared to virgin refining).
- **Reduced Hazardous Waste:** Unlike acid-clay refining, hydrotreating avoids the generation of acid sludge, thus reducing toxic waste disposal costs.
- **Economic Viability:** The operating cost of refining used oil is generally 30%–50% lower than producing base oil from virgin crude.
- **Versatile Feedstock Processing:** Modern plants can process varied, contaminated, and low-grade waste oil into standardized, clean products.

➢ Weaknesses (Internal)

- **High Capital Expenditure (CAPEX):** Setting up a full re-refining plant with hydroprocessing can exceed $150 million.
- **Technical Complexity & High Operating Costs (OPEX):** Hydrotreatment is a complex, high-pressure, high-temperature process requiring skilled manpower and specialized, expensive catalysts.
- **Sensitivity to Feedstock Quality:** While advanced, the technology can still be affected by severely contaminated, mixed waste streams (e.g., oil mixed with solvents or PCBs).
- **Lower Initial Margin for Smaller Plants:** Without economies of scale, smaller operations may struggle with high depreciation and utility costs.

➢ Opportunities (External)

- **Stringent Environmental Regulations:** Globally, over 60 countries have tightened regulations on waste oil disposal, forcing a shift from burning to recycling, creating a surge in demand for, authorized, high-tech recycling services.
- **Growing Circular Economy Market:** The global circular economy for lubricants is projected to grow significantly, offering long-term contracts with industrial users seeking green, certified products.
- **Import Substitution:** Countries with high demand for lubricant base stocks (like India or in Africa) can replace costly imports with domestic re-refined oil.
- **Supportive Policies and Subsidies:** Many governments provide incentives, tax waivers, and grants for projects that align with green energy and sustainable development goals.
- **Technological Advancements:** Increased use of AI and IoT for smart, automated waste collection and improved, higher-yield refining processes (e.g., thin-film evaporation).

➢ Threats (External)

- **Feedstock Collection Logistics:** Securing a consistent, clean supply of used oil is difficult due to unauthorized dumping and competition with low-cost, illegal burners.
- **Price Volatility of Crude Oil:** A sharp drop in virgin oil prices can make re-refined oil less economically competitive.

- **Competition from Conventional Burning:** In regions with weak enforcement, used oil is still favored as a cheap, direct industrial fuel, making it difficult to collect for refining.
- **Regulatory Inconsistencies:** Disparities in local, national, and regional policies can lead to inconsistent compliance standards and unfair competition.
- **Technical Failure/Contamination Risks:** High-value hydrotreaters are sensitive to specialized contamination, which could halt production or ruin catalysts.

CAPITAL REQUIUREMENT

Cost of project

Amount in Lacs

Particular	Amount (Basic)	GST Over Capital Assets	Project Cost with GST	Interest to be Capitalized		Total Project Cost (with Interest)
				Interest - Yr 1	Interest - Yr 2	
Land Cost	1,200	-	1,200	-	-	1,200
Civil Work	1,620	292	1,912	37.17	97.39	2,046
Distilation & Hydro Treatment Plant	6,091	1,096	7,187	-	226.40	7,414
Other Equipment and Machinery	1,621	292	1,913	-	60.25	1,973
Elecrtrical Installation	450	81	531	-	16.73	548
DG Set	195	35	230	-	7.25	237
Other Fixed Assets	482	87	569	-	-	569
Total Project cost	**11,659**	**1,883**	**13,542**	**37**	**408**	**13,987**
Working Capital	2,000	-	2,000	-	-	2,000
Total Working Capital	**2,000**	**-**	**2,000**	**-**	**-**	**2,000**
TOTAL FUND Requirement	**13,659**	**1,883**	**15,542**	**37**	**408**	**15,987**

Sources of Funds

Amount in Lacs

Particulars	% of Share	Amount
Promotors Contribution	30%	4,662
Term Loan for Plant	70%	9,479
Working Capital Loan	70%	1,400
Total Sources of Funds		**15,542**

-

Projected Balance Sheet

Amount in Lacs

Liabilities	Pre-Operation Period		Post Operation Period				
	Year 1 (6 Months)	*Year 2*	*Year 3*	*Year 4*	*Year 5*	*Year 6*	*Year 7*
Capital	714.00	3,891.86	4,662.49	4,662.49	4,662.49	4,662.49	4,662.49
Reserves & Surplus	-	-	-522.26	729.79	2,138.95	3,875.77	5,915.83
Term Loan	1,740.97	9,646.33	9,444.47	7,694.47	5,694.47	3,494.47	-
Working Capital Loan	-	-	1,400.00	1,400.00	1,400.00	1,400.00	1,400.00
Trades Payables	-	-	9.86	9.91	9.96	10.01	10.06
Other Current Liabilities	15.00	30.00	60.95	164.23	175.72	184.08	192.96
Total	**2,469.97**	**13,568.19**	**15,055.50**	**14,660.89**	**14,081.57**	**13,626.81**	**12,181.34**

Assets	Pre-Operation Period		Post Operation Period				
	Year 1 (6 Months)	*Year 2*	*Year 3*	*Year 4*	*Year 5*	*Year 6*	*Year 7*
Fixed Assets	1,200.00	1,200.00	12,395.79	12,395.79	12,395.79	12,395.79	12,395.79
CWIP	1,217.17	10,713.79	-	-	-	-	-
Depreciation	-	-	1,590.56	2,949.17	4,252.77	5,367.27	6,320.70
Net Block	***2,417.17***	***11,913.79***	***10,805.23***	***9,446.62***	***8,143.02***	***7,028.52***	***6,075.08***
Short Term Investment	-	-	1,200.00	2,450.00	2,950.00	3,700.00	3,200.00
Deferred Expenses	52.80	150.15	120.12	90.09	60.06	30.03	-
Trade Receivables	-	-	764.51	1,136.43	1,172.45	1,202.02	1,232.35
Inventory	-	-	490.73	718.57	736.54	754.95	773.83
Other Current Assets	-	1,504.26	1,413.62	697.17	705.12	722.75	740.82
Cash & Bank Balance	-	-	261.30	122.01	314.39	188.53	159.26
Total	**2,469.97**	**13,568.19**	**15,055.50**	**14,660.89**	**14,081.57**	**13,626.81**	**12,181.34**

Projected Profit & Loss Statament

Amount in Lacs

Particulars	Pre-Operation Period		Post Operation Period				
	Year 1 (6 Months)	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7
Revenue							
Fuel	-	-	1,119	1,663	1,716	1,759	1,803
Base Oil - (Lubricant)	-	-	8,954	13,308	13,726	14,070	14,421
Wastage (with Salvage Value)	-	-	681	1,011	1,043	1,069	1,096
Frieght Revenue	-	-	507	757	784	808	832
Other Income	-	-	54	164	243	299	333
Total	**-**	**-**	**11,315**	**16,903**	**17,513**	**18,005**	**18,485**
Expenses							
Cost of Goods Sold	-	-	6,859	10,036	10,284	10,541	10,804
Consumables & Spares	-	-	240	241	242	244	245
Other Production Cost	-	-	17	18	20	21	23
Repair & Maint	-	-	112	113	113	114	114
Electricity & Power	-	-	483	710	731	752	774
Transportation Cost	-	-	637	909	924	952	980
Employee Expenses	-	-	731	790	853	921	995
Staff Training & Welfare	-	-	15	16	17	18	20
Admin Expenses	-	-	37	39	43	46	50
Marketing Expenses	-	-	56	84	86	89	91
Finance Expenses	-	-	1,030	976	819	639	441
Amortization of Deferred Exps	-	-	30	30	30	30	30
Depreciation	-	-	1,591	1,359	1,304	1,114	953
Total	**-**	**-**	**11,837**	**15,320**	**15,465**	**15,480**	**15,520**
Profit before Tax	-	-	-522	1,583	2,048	2,524	2,965
Income Tax @ 31.2%	-	-	-	331	639	788	925
Profit after Tax	-	-	-522	1,252	1,409	1,737	2,040
Cash Profit (Profit before Dep)			1,098	2,641	2,743	2,881	3,024

Reserves & Surplus **Amount in Lacs**

Particulars	*Pre-Operation Period*		*Post Operation Period*				
	Year 1 (6 Months)	*Year 2*	*Year 3*	*Year 4*	*Year 5*	*Year 6*	*Year 7*
Opening Reserves	-	-	-	-522	730	2,139	3,876
Addition	-	-	-522	1,252	1,409	1,737	2,040
Utilization/Reduction	-	-	-	-	-	-	-
Closing Reserves	**-**	**-**	**-522**	**730**	**2,139**	**3,876**	**5,916**

Tax Calculation

Particulars	*Pre-Operation Period*		*Post Operation Period*				
	Year 1 (6 Months)	*Year 2*	*Year 3*	*Year 4*	*Year 5*	*Year 6*	*Year 7*
Net Loss Cf	-	-	-	-522	-	-	-
Current Year Profit	-	-	-522	1,583	2,048	2,524	2,965
Net Taxable Amount / Loss CF	-	-	-522	1,061	2,048	2,524	2,965
Tax Liability @ 31.2%	-	-		331	639	788	925

GP Ratio	26%	28%	29%	29%	29%
NP Ratio - Beftore Tax & Dep	9%	17%	19%	20%	21%
NP Ratio - Beftore Tax	-5%	9%	12%	14%	16%
NP Ratio - After Tax	-5%	7%	8%	10%	11%
GP	2,967	4,877	5,199	5,382	5,545
NP - Beftore Tax & Dep	1,068	2,942	3,352	3,639	3,919
NP - Beftore Tax	-522	1,583	2,048	2,524	2,965
NP - After Tax	-522	1,252	1,409	1,737	2,040

CASH FLOW STATEMENT

Amount in Lacs

Particulars	*Pre-Operation Period*		*Post Operation Period*				
	Year 1 (6 Months)	*Year 2*	*Year 3*	*Year 4*	*Year 5*	*Year 6*	*Year 7*
Profit before Tax	-	-	-522.26	1,583.01	2,048.19	2,524.45	2,965.20
Add:							
Dep	-	-	1,590.56	1,358.61	1,303.60	1,114.50	953.44
Trade Receivable	-	-	-764.51	-371.92	-36.02	-29.58	-30.32
Inventory	-	-	-490.73	-227.84	-17.96	-18.41	-18.87
Short Term Investment	-	-	-1,200.00	-1,250.00	-500.00	-750.00	500.00
Deferred Expenditure	-52.80	-97.35	30.03	30.03	30.03	30.03	30.03
Other Current Assets	-	-1,504.26	90.64	716.45	-7.96	-17.63	-18.07
Trade Payables	-	-	9.86	0.05	0.05	0.05	0.05
Working Capital Loan (CC/OD)	-	-	1,400.00	-	-	-	-
Other Current Liabities	15.00	15.00	30.95	103.28	11.48	8.36	8.89
Cash from Operating Activities	**-37.80**	**-1,586.61**	**174.54**	**1,941.66**	**2,831.41**	**2,861.78**	**4,390.34**
Less: Income Tax	-	-	-	-330.95	-639.04	-787.63	-925.14
Net Cash flow from Operating Activities	**-37.80**	**-1,586.61**	**174.54**	**1,610.71**	**2,192.38**	**2,074.15**	**3,465.19**
Cash Flow from Investing							
FA - (addition)/Sale of FA	-1,200.00	-	-11,195.79	-	-	-	-
CWIP	-1,217.17	-9,496.62	10,713.79	-	-	-	-
Cash Flow from Financing							
Capital Contribution	714.00	3,177.86	770.63	-	-	-	-
Debt - Increase / (Repayment)	1,740.97	7,905.36	-201.87	-1,750.00	-2,000.00	-2,200.00	-3,494.47
Cash & Bank Balance							
Opening	-	-	-	261.30	122.01	314.39	188.53
Closing	-	-	261.30	122.01	314.39	188.53	159.26

FINANCE COST

Amount in Lacs

Particulars	Pre-Operation Period		Post Operation Period				
	Year 1 (6 Months)	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7
Opening Balance	-	1,741	9,646	9,444	7,694	5,694	3,494
Addition (assumed at the first date of the Year)	1,666	7,415	398	-	-	-	-
Addition - Interest	75	490	904	850	693	513	315
Repayment							
Interest	-	-	904	850	693	513	315
Principal	-	-	600	1,750	2,000	2,200	3,494
Closing Balance	1,741	9,646	9,444	7,694	5,694	3,494	-
Interest Rate - Term Loan	9%	9%	9%	9%	9%	9%	9%
Finance Cost Capitalized	**37**	**408**	**-**	**-**	**-**	**-**	**-**
Finance Cost charged to PL (Term Loan)	**38**	**82**	**904**	**850**	**693**	**513**	**315**
TOTAL INTEREST CHARGED TO PL	**38**	**82**	**1,030**	**976**	**819**	**639**	**441**

Working Capital

Amount in Lacs

Particulars	Pre-Operation Period		Post Operation Period				
	Year 1 (6 Months)	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7
Opening Balance	-	-	-	1,400	1,400	1,400	1,400
Addition (Initial date of the Year)	-	-	1,400	-	-	-	-
Repayment	-	-	-	-	-	-	-
Closing Balance	-	-	1,400	1,400	1,400	1,400	1,400
Interest Rate - Working Capital Loan	9%	9%	9%	9%	9%	9%	9%
Interest - Working Capital	**-**	**-**	**126**	**126**	**126**	**126**	**126**

NOTES - BS

Notes - Trade Payables — **Amount in Lacs**

Particulars	*Pre-Operation Period*		*Post Operation Period*				
	Year 1 (6 Months)	*Year 2*	*Year 3*	*Year 4*	*Year 5*	*Year 6*	*Year 7*
Trade Payables			9.86	9.91	9.96	10.01	10.06

Notes - Other Current Liabilites — **Amount in Lacs**

Particulars	*Pre-Operation Period*		*Post Operation Period*				
	Year 1 (6 Months)	*Year 2*	*Year 3*	*Year 4*	*Year 5*	*Year 6*	*Year 7*
Salary Payables	-	-	60.95	65.83	71.09	76.78	82.92
Deferred Exps - Admin Expenses	15.00	30.00	-	-	-	-	-
GST Liabilities	-	-	-	98.41	104.62	107.30	110.04
Provision for Income Tax	-	-	-	-	-	-	-
Total Other Current Liablities	**15**	**30**	**61**	**164**	**176**	**184**	**193**

Notes - Short Term Investment — **Amount in Lacs**

Particulars	*Pre-Operation Period*		*Post Operation Period*				
	Year 1 (6 Months)	*Year 2*	*Year 3*	*Year 4*	*Year 5*	*Year 6*	*Year 7*
Opening	-	-	-	1,200	2,450	2,950	3,700
Addition	-	-	1,200	1,250	500	750	-
Reduction	-	-	-	-	-	-	500
Closing Balance	**-**	**-**	**1,200**	**2,450**	**2,950**	**3,700**	**3,200**
Interest Rate	9%	9%	9%	9%	9%	9%	9%
Interest	**-**	**-**	**54**	**164**	**243**	**299**	**333**

NOTES - BS

Notes - Deferred Expenditure **Amount in Lacs**

Particulars	*Pre-Operation Period*		*Post Operation Period*				
	Year 1 (6 Months)	*Year 2*	*Year 3*	*Year 4*	*Year 5*	*Year 6*	*Year 7*
Opening	-	53	150	120	90	60	30
Addition							
Admin Expenses	15	15	-	-	-	-	-
Interest	38	82	-	-	-	-	-
Less: Charged to PL	-	-	30	30	30	30	30
Closing Balance	***53***	***150***	***120***	***90***	***60***	***30***	***-***

Notes - Debtors **Amount in Lacs**

Particulars	*Pre-Operation Period*		*Post Operation Period*				
	Year 1 (6 Months)	*Year 2*	*Year 3*	*Year 4*	*Year 5*	*Year 6*	*Year 7*
Debetors							
Sales Credit Days	21	21	21	21	21	21	21
Total Annual Sales Net of GST	-	-	11,260.90	16,739.19	17,269.71	17,705.37	18,152.04
Closing Debtors (Net of GST)	-	-	647.89	963.08	993.60	1,018.67	1,044.36
Add: GST over Closing Debtors	-	-	116.62	173.35	178.85	183.36	187.99
Closing Debtors	**-**	**-**	**764.51**	**1,136.43**	**1,172.45**	**1,202.02**	**1,232.35**

NOTES - BS

Notes - Closing Stock **Amount in Lacs**

Particulars	*Pre-Operation Period*		*Post Operation Period*				
	Year 1 (6 Months)	*Year 2*	*Year 3*	*Year 4*	*Year 5*	*Year 6*	*Year 7*
Closing Stock - Black OIL	-	-	307.13	449.72	460.96	472.49	484.30
Closing Stock - Processed Oil - (FG)	-	-	183.61	268.86	275.58	282.47	289.53
Total Closing Stock	**-**	**-**	**490.73**	**718.57**	**736.54**	**754.95**	**773.83**

Notes - Other Current Assets **Amount in Lacs**

Particulars	*Pre-Operation Period*		*Post Operation Period*				
	Year 1 (6 Months)	*Year 2*	*Year 3*	*Year 4*	*Year 5*	*Year 6*	*Year 7*
Advance to Trades Payable							
Annual Purchases	-	-	7,166.25	10,178.63	10,294.81	10,552.18	10,815.99
Advance to Supplier	-	-	490.84	697.17	705.12	722.75	740.82
(Advance 25 Days of Purchases)							
GST - Input	-	1,504.26	922.78	-	-	-	-
Total Other Current Assets	**-**	**1,504.26**	**1,413.62**	**697.17**	**705.12**	**722.75**	**740.82**

NOTES - PL

Notes - Consumables, Spares & Repair Maint — **Amount in Lacs**

Particulars	*Pre-Operation Period*		*Post Operation Period*				
	Year 1 (6 Months)	*Year 2*	*Year 3*	*Year 4*	*Year 5*	*Year 6*	*Year 7*
P&M - Gross Value	-	-	7,998.66				
3% of P&M	-	-	239.96	241.16	242.37	243.58	244.79

Notes - ETP — **Amount in Lacs**

Particulars	*Pre-Operation Period*		*Post Operation Period*				
	Year 1 (6 Months)	*Year 2*	*Year 3*	*Year 4*	*Year 5*	*Year 6*	*Year 7*
ETP Cost per Day			5000				
Annual Working Day			335				
Total Annual ETP Cost	**-**	**-**	**1,675,000**	**1,809,000**	**1,953,720**	**2,110,018**	**2,278,819**

Notes - Repair & Maintenance — **Amount in Lacs**

Particulars	*Pre-Operation Period*		*Post Operation Period*				
	Year 1 (6 Months)	*Year 2*	*Year 3*	*Year 4*	*Year 5*	*Year 6*	*Year 7*
Building/P&M/Other Equipment/Vehicle-Gross Value	-	-	11,195.79				
1% of P&M and Other Eqipment / Vehicle	-	-	111.96	112.52	113.08	113.65	114.21

NOTES - PL

Notes - Electricity Cost **Amount in INR**

Particulars	*Pre-Operation Period*		*Post Operation Period*				
	Year 1 (6 Months)	*Year 2*	*Year 3*	*Year 4*	*Year 5*	*Year 6*	*Year 7*
100 TPD Electricity Consumption - Per Hour (KV)			800	800	800	800	800
Electricity Consumption - Per hour (KV)			364	520	520	520	520
Daily Working Hours			18	18	18	18	18
Daily Consumption (KV)			6,552	9,360	9,360	9,360	9,360
Working Days			335	335	335	335	335
Annual Consumption (KV)			2,194,920	3,135,600	3,135,600	3,135,600	3,135,600
Rate per Unit - per KV			12	12.24	12.48	12.73	12.99
Annual Electricity Cost (In INR)	-	-	***26,339,040***	***38,379,744***	***39,147,339***	***39,930,286***	***40,728,891***
DG Set Operating Expenses							
Operating Hours per Day			6	6	6	6	6
Unit Consumption per Hour			364	520	520	520	520
Daily Consumption - in Units			2,184	3,120	3,120	3,120	3,120
DG Cost per Unit			30.00	31.20	32.45	33.75	35.10
DG Cost Per Day			65,520	97,344	101,238	105,287	109,499
Working Day			335	335	335	335	335
Annual DG Cost (In INR)	-	-	***21,949,200***	***32,610,240***	***33,914,650***	***35,271,236***	***36,682,085***
TOTAL Electricity & DG Set (in INR)			**48,288,240**	**70,989,984**	**73,061,988**	**75,201,521**	**77,410,976**

NOTES - PL

Notes - Transportation Cost — **Amount in INR**

Particulars	*Pre-Operation Period*		*Post Operation Period*				
	Year 1 (6 Months)	*Year 2*	*Year 3*	*Year 4*	*Year 5*	*Year 6*	*Year 7*
Total Purchase of Black Oil (in KL)	-	-	15,925	22,068	21,775	21,775	21,775
Tranportation Cost for per Ltr	-	-	4	4	4	4	5
Tranportation Cost for per KL			4,000	4,120	4,244	4,371	4,502
Total Transportation Cost - Annual (In INR)			***63,700,000***	***90,918,100***	***92,404,390***	***95,176,522***	***98,031,817***

Notes - Staff Training & Welfare Cost — **Amount in Lacs**

Particulars	*Pre-Operation Period*		*Post Operation Period*				
	Year 1 (6 Months)	*Year 2*	*Year 3*	*Year 4*	*Year 5*	*Year 6*	*Year 7*
Employee Cost	-	-	731.40	789.91	853.10	921.35	995.06
Staff Training & Welfare Cost @ 2% of Employee C	-	-	14.63	15.80	17.06	18.43	19.90

Notes - Admin Cost — **Amount in Lacs**

Particulars	*Pre-Operation Period*		*Post Operation Period*				
	Year 1 (6 Months)	*Year 2*	*Year 3*	*Year 4*	*Year 5*	*Year 6*	*Year 7*
Employee Cost	-	-	731.40	789.91	853.10	921.35	995.06
Admin Cost @ 5% of Employee Cost	-	-	36.57	39.50	42.66	46.07	49.75

Notes - Marketing Expenses — **Amount in Lacs**

Particulars	*Pre-Operation Period*		*Post Operation Period*				
	Year 1 (6 Months)	*Year 2*	*Year 3*	*Year 4*	*Year 5*	*Year 6*	*Year 7*
Sales	-	-	11,260.90	16,739.19	17,269.71	17,705.37	18,152.04
0.5% of Sales	-	-	56.30	83.70	86.35	88.53	90.76

REVENUE, COGS & STOCK

Revenue — **Amount in INR**

Particulars	*Pre-Operation Period*			*Post Operation Period*				
	Year 1 (6 Months)	*Year 2*		*Year 3*	*Year 4*	*Year 5*	*Year 6*	*Year 7*
Daily Processing (in KL)	-	-		46	65	65	65	65
No of Annual Working Day	-	-		335	335	335	335	335
Total Annual Processing (Black OIL) (KL)				15,243	21,775	21,775	21,775	21,775
Production in KL (Annual)			**OutPut**					
Fuel	-	-	0.10	1,524	2,178	2,178	2,178	2,178
Base Oil - (Lubricant)	-	-	0.75	11,432	16,331	16,331	16,331	16,331
Wastage (with Salvage Value)	-	-	0.12	1,829	2,613	2,613	2,613	2,613
Evaporation	-	-	0.03	457	653	653	653	653
Clsoing Stock - Processed Oil (FG) (in KL)								
Assumed - 7 Days Stock								
Fuel	-	-		32	46	46	46	46
Base Oil - (Lubricant)	-	-		239	341	341	341	341
Wastage (with Salvage Value)	-	-		38	55	55	55	55
Total Sales - Processed OIL (in KL) (Annual)								
Fuel				1,492	2,164	2,178	2,178	2,178
Base Oil - (Lubricant)				11,193	16,229	16,331	16,331	16,331
Wastage (with Salvage Value)				1,791	2,597	2,613	2,613	2,613
Revenue								
Sales Price per KL								
Fuel				75,000	76,875	78,797	80,767	82,786
RRBO (Re-refine Base Oil API Group II+)				80,000	82,000	84,050	86,151	88,305
Wastage (with Salvage Value)				38,000	38,950	39,924	40,922	41,945
Total Annual Revenue								
Fuel				111,930,000	166,345,969	171,580,195	175,869,700	180,266,443
RRBO (Re-refine Base Oil API Group II+)				895,440,000	1,330,767,750	1,372,641,563	1,406,957,602	1,442,131,542
Wastage (with Salvage Value)				68,053,440	101,138,349	104,320,759	106,928,778	109,601,997
Freight - Revenue				50,666,980	75,666,589	78,428,226	80,781,073	83,204,505
TOTAL REVENUE				**1,126,090,420**	**1,673,918,656**	**1,726,970,743**	**1,770,537,152**	**1,815,204,486**

REVENUE, COGS & STOCK

Freight Revenue **Amount in INR**

Particulars	*Pre-Operation Period*		*Post Operation Period*				
	Year 1 (6 Months)	*Year 2*	*Year 3*	*Year 4*	*Year 5*	*Year 6*	*Year 7*
Freight (Recovery) - Revenue per Ltr	-	-					
Freight (Recovery) - Revenue per KL	-	-	3,500.00	3,605.00	3,713.15	3,824.54	3,939.28
Total Sales - Annual (in KL)	-	-	14,476	20,989	21,122	21,122	21,122
Total Freight - Revenue - Annuall			***50,666,980***	***75,666,589***	***78,428,226***	***80,781,073***	***83,204,505***

Feed Stock - (Used Black Oil / Lubricant)
Working for PURCAHSE, COGS & Stock **Amount in INR**

Particulars	*Pre-Operation Period*		*Post Operation Period*				
	Year 1 (6 Months)	*Year 2*	*Year 3*	*Year 4*	*Year 5*	*Year 6*	*Year 7*
Black Oil Annual Purchase & Stock (in KL)							
No of Annual Working Day	-	-	335	335	335	335	335
Annual Processing (in KL)	-	-	15,243	21,775	21,775	21,775	21,775
Closing Stock (***15 Days***) (***in KL***)	-	-	683	975	975	975	975
Opening Stock (in KL)	-	-	-	683	975	975	975
Annual Purchase - (in KL)	-	-	15,925	22,068	21,775	21,775	21,775
Purchase Cost per KL	-	-	45,000	46,125	47,278	48,460	49,672
Annual Purchase cost (in INR)	-	-	716,625,000	1,017,863,438	1,029,481,172	1,055,218,201	1,081,598,656
Opening Stock	-	-	-	30,712,500	44,971,875	46,096,172	47,248,576
Closing Stock (in INR)	-	-	30,712,500	44,971,875	46,096,172	47,248,576	48,429,791
Cost of Goods Sold - Annual			**685,912,500**	**1,003,604,063**	**1,028,356,875**	**1,054,065,797**	**1,080,417,442**

REVENUE, COGS & STOCK

Closing Stock of Processed OIL (FG) **Amount in INR**

Particulars	*Pre-Operation Period*		*Post Operation Period*				
	Year 1 (6 Months)	*Year 2*	*Year 3*	*Year 4*	*Year 5*	*Year 6*	*Year 7*
Closing Stock of Processed OIL (FG) (in KL)							
Fuel	-	-	32	46	46	46	46
Base Oil - (Lubricant)	-	-	239	341	341	341	341
Wastage (with Salvage Value)	-	-	38	55	55	55	55
Cost Per KL							
(Assumed Cost of FG shall be 80% of Sales Price)							
Fuel			60,000	61,500	63,038	64,613	66,229
Base Oil - (Lubricant)			64,000	65,600	67,240	68,921	70,644
Wastage (with Salvage Value)			30,400	31,160	31,939	32,737	33,556
Closing Stock Value							
Fuel			1,911,000	2,798,250	2,868,206	2,939,911	3,013,409
Base Oil - (Lubricant)			15,288,000	22,386,000	22,945,650	23,519,291	24,107,274
Wastage (with Salvage Value)			1,161,888	1,701,336	1,743,869	1,787,466	1,832,153
Total Value of Closing Stock			**18,360,888**	**26,885,586**	**27,557,726**	**28,246,669**	**28,952,836**

EMPLOYEE COST

Amount in INR

MAN POWER	Nos	Salary per Month	Annual Cost	Year 3	Year 4	Year 5	Year 6	Year 7
Plant Manager (Incharge)	1	325,000	3,900,000	3,900,000	4,212,000	4,548,960	4,912,877	5,305,907
Process Engineer - Operation	4	120,000	5,760,000	5,760,000	6,220,800	6,718,464	7,255,941	7,836,416
Mechanincal - Engineer	2	120,000	2,880,000	2,880,000	3,110,400	3,359,232	3,627,971	3,918,208
Electrical	2	120,000	2,880,000	2,880,000	3,110,400	3,359,232	3,627,971	3,918,208
Instrumentation	1	120,000	1,440,000	1,440,000	1,555,200	1,679,616	1,813,985	1,959,104
Operator	8	60,000	5,760,000	5,760,000	6,220,800	6,718,464	7,255,941	7,836,416
WeighBridge - Operator	2	30,000	720,000	720,000	777,600	839,808	906,993	979,552
Helper	20	30,000	7,200,000	7,200,000	7,776,000	8,398,080	9,069,926	9,795,521
LAB								
Lab - Incharge	1	120,000	1,440,000	1,440,000	1,555,200	1,679,616	1,813,985	1,959,104
Technician	3	50,000	1,800,000	1,800,000	1,944,000	2,099,520	2,267,482	2,448,880
Helper	3	30,000	1,080,000	1,080,000	1,166,400	1,259,712	1,360,489	1,469,328
Support Staff								
Accounts / Finance	3	150,000	5,400,000	5,400,000	5,832,000	6,298,560	6,802,445	7,346,640
Procurement - Incharge	1	60,000	720,000	720,000	777,600	839,808	906,993	979,552
Procurement - Officer	1	30,000	360,000	360,000	388,800	419,904	453,496	489,776
Logistic	3	35,000	1,260,000	1,260,000	1,360,800	1,469,664	1,587,237	1,714,216
Stores	1	25,000	300,000	300,000	324,000	349,920	377,914	408,147
Safety								
Safety Incharge	1	120,000	1,440,000	1,440,000	1,555,200	1,679,616	1,813,985	1,959,104
Safety Officer	2	80,000	1,920,000	1,920,000	2,073,600	2,239,488	2,418,647	2,612,139
Safety - Support	6	35,000	2,520,000	2,520,000	2,721,600	2,939,328	3,174,474	3,428,432
Guards - Supervisor	2	50,000	1,200,000	1,200,000	1,296,000	1,399,680	1,511,654	1,632,587
Guards	16	30,000	5,760,000	5,760,000	6,220,800	6,718,464	7,255,941	7,836,416
Marketing								
Marketing - Incharge	1	180,000	2,160,000	2,160,000	2,332,800	2,519,424	2,720,978	2,938,656
Marketing - Support	2	70,000	1,680,000	1,680,000	1,814,400	1,959,552	2,116,316	2,285,621
Driver								
Driver - Tanker - 2 No	2	30,000	720,000	720,000	777,600	839,808	906,993	979,552
Helper - Tanker - 2 No	2	30,000	720,000	720,000	777,600	839,808	906,993	979,552
Driver - Truck - 3 No	3	20,000	720,000	720,000	777,600	839,808	906,993	979,552
Helper - Truck - 3 No	3	20,000	720,000	720,000	777,600	839,808	906,993	979,552
Driver - Pick up 2 No	2	30,000	720,000	720,000	777,600	839,808	906,993	979,552
Driver - Lift & Tractor	2	30,000	720,000	720,000	777,600	839,808	906,993	979,552
Driver - Car 2 Nos	2	30,000	720,000	720,000	777,600	839,808	906,993	979,552
Driver - Force	1	30,000	360,000	360,000	388,800	419,904	453,496	489,776
Ambulance - 1 Nos	1	30,000	360,000	360,000	388,800	419,904	453,496	489,776
Pantry / Guest House	3	25,000	900,000	900,000	972,000	1,049,760	1,133,741	1,224,440
HR / Admin / Legal	1	120,000	1,440,000	1,440,000	1,555,200	1,679,616	1,813,985	1,959,104
Hr - Support	1	30,000	360,000	360,000	388,800	419,904	453,496	489,776
Sweeper	4	25,000	1,200,000	1,200,000	1,296,000	1,399,680	1,511,654	1,632,587
Gardener	3	25,000	900,000	900,000	972,000	1,049,760	1,133,741	1,224,440
Medic - Staff	2	50,000	1,200,000	1,200,000	1,296,000	1,399,680	1,511,654	1,632,587
Misc Salary	1	150,000	1,800,000	1,800,000	1,944,000	2,099,520	2,267,482	2,448,880
Total	**119**	**2,665,000**	**73,140,000**	**73,140,000**	**78,991,200**	**85,310,496**	**92,135,336**	**99,506,163**
Monthly Salary			**6,095,000**	**6,095,000**	**6,582,600**	**7,109,208**	**7,677,945**	**8,292,180**

DEPRECIATION

Amount in Lacs

Particulars	Purchase Cost	GST Capitalization	Interest to Be Capitailsed	Total Gross Block	Rate of Dep	Depreciation					Closing WDV				
						Year 3	Year 4	Year 5	Year 6	Year 7	Year 3	Year 4	Year 5	Year 6	Year 7
Land Cost	1,200	-	-	**1,200**	0%	-	-	-	-	-	1,200	1,200	1,200	1,200	1,200
Civil Work															
Civil Work	1,500	270	135	**1,905**	10%	190	171	154	139	125	1,714	1,543	1,388	1,250	1,125
Roads	35	6	-	**41**	10%	4	4	3	3	3	37	33	30	27	24
Storage Area	35	6	-	**41**	10%	4	4	3	3	3	37	33	30	27	24
Streat Light	25	5	-	**30**	10%	3	3	2	2	2	27	24	22	19	17
Painting	25	5	-	**30**	10%	3	3	2	2	2	27	24	22	19	17
Equipment and Machinery															
Distilation	3,280	-	-	**3,280**	15%	492	418	429	365	310	2,788	2,862	2,433	2,068	1,758
Hydro Treatment Plant	2,811	-	226	**3,037**	15%	456	387	398	338	287	2,582	2,650	2,253	1,915	1,628
Fractionation	206	-	60	**266**	15%	40	34	29	25	21	226	192	164	139	118
Laboratory & Instruments	185	-	-	**185**	15%	28	24	20	17	14	157	134	114	97	82
Fire Proofing	160	-	-	**160**	15%	24	20	17	15	13	136	116	98	84	71
Chemicals & Oil	70	-	-	**70**	15%	11	9	8	6	5	60	51	43	37	31
Misc	1,000	-	-	**1,000**	15%	150	128	108	92	78	850	723	614	522	444
Elecrtrical Installation															
Electrical OSBL (outside Battery Limit)	450	-	17	**467**	15%	70	60	51	43	37	397	337	287	244	207
DG - 550 KVA (3 No)	195	-	7	**202**	15%	30	26	22	19	16	172	146	124	106	90
Office Equipment															
Office Equipment	150	-	-	**150**	20%	30	24	19	15	12	120	96	77	61	49
Computer	30	-	-	**30**	35%	11	7	4	3	2	20	13	8	5	3
Vehicle															
Vehicle - Tanker - 2 No	100	-	-	**100**	15%	15	13	11	9	8	85	72	61	52	44
Vehicle - Truck - 3 No	75	-	-	**75**	15%	11	10	8	7	6	64	54	46	39	33
Vehicle - Pick up 2 No	30	-	-	**30**	15%	5	4	3	3	2	26	22	18	16	13
Lift & Tractor	12	-	-	**12**	15%	2	2	1	1	1	10	9	7	6	5
Car 2 Nos	20	-	-	**20**	15%	3	3	2	2	2	17	14	12	10	9
Force - 20 Seater (1 No)	20	-	-	**20**	15%	3	3	2	2	2	17	14	12	10	9
Ambulances (1 No)	10	-	-	**10**	15%	2	1	1	1	1	9	7	6	5	4
Fire Tender	35	-	-	**35**	15%	5	4	4	3	3	30	25	21	18	16
TOTAL	**11,659**	**292**	**445**	**12,396**		**1,591**	**1,359**	**1,304**	**1,114**	**953**	**10,805**	**10,394**	**9,091**	**7,976**	**7,023**